

- 3) Use a screwdriver to unhook the locking lever of the right locking plate and pull the locking plate off of the bus module.

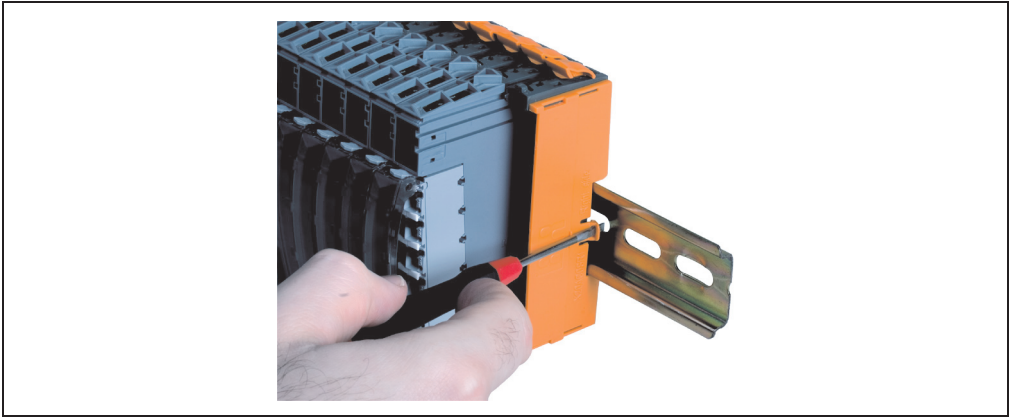


Figure 215: Unlock the right locking plate with a screwdriver

- 4) Now you can add more modules as described for assembly method 2 (see section 2.2 "Method 2" on page 430).

6. Installation of accessories

6.1 Additional locking mechanisms

Some specific areas require additional locking mechanisms to prevent accidental release of the mechanics.

6.1.1 Additional locking clip

The additional clip attaches the electronic module to the bus module. The locking clip is inserted in the appropriate opening on the module and pushed down.

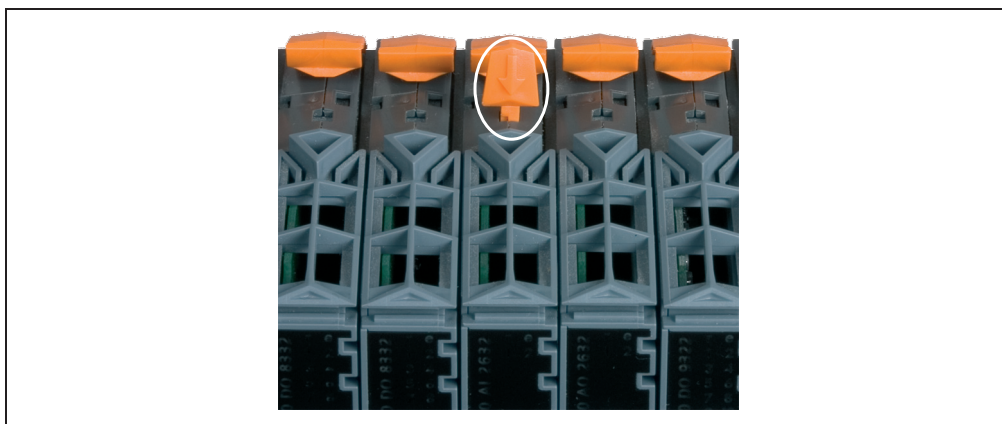


Figure 216: Installing the additional locking clip

6.1.2 Terminal locking clip

The terminal locking clip attaches the terminal block securely to the electronic module.

- 1) Set the terminal locking clip on the terminal block locking lever as shown.

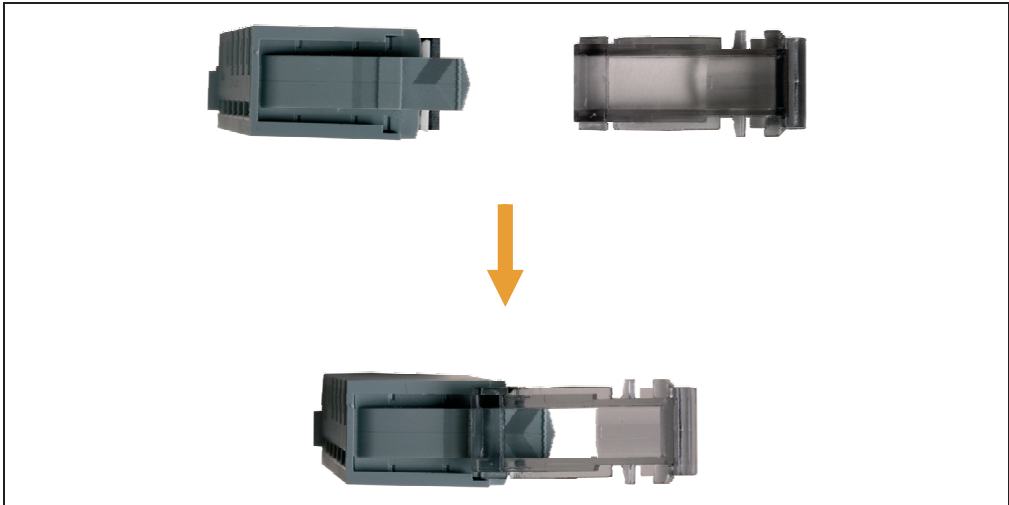


Figure 217: Set the terminal locking clip on the terminal block locking lever

- 2) Push down and hold the terminal locking clip and the locking lever with your index finger ①. Finally, slide the terminal locking clip forward with your thumb ②.

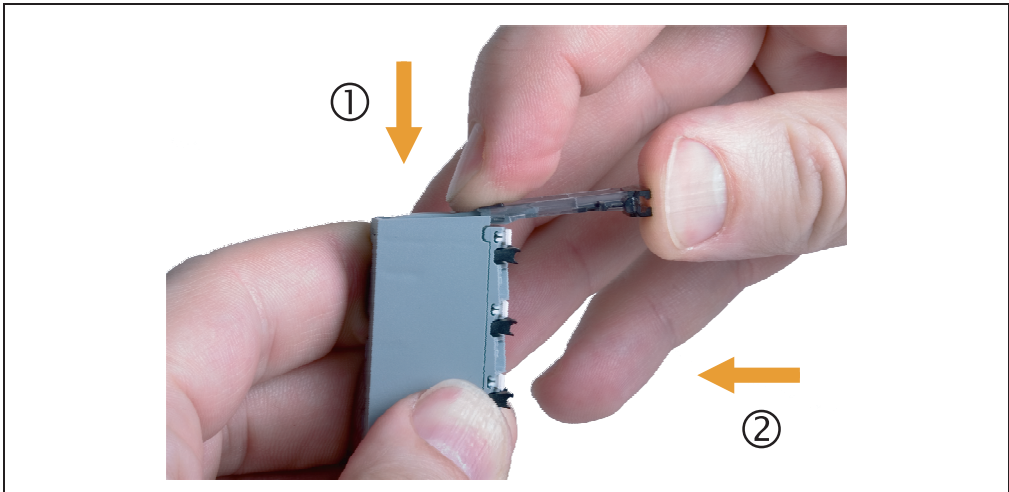


Figure 218: Install terminal locking clip on terminal block

- 3) Hang the bottom edge of the terminal block in its place on the bus module.

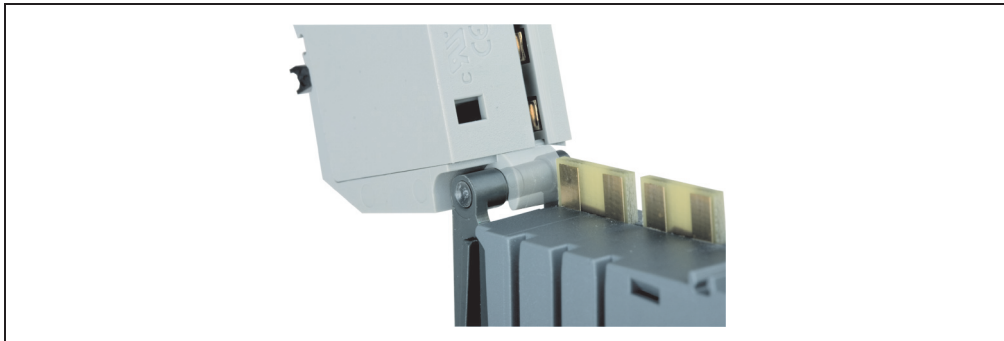


Figure 219: Hang the bottom edge of the terminal block in its place on the bus module

- 4) Rotate the terminal block up into place.

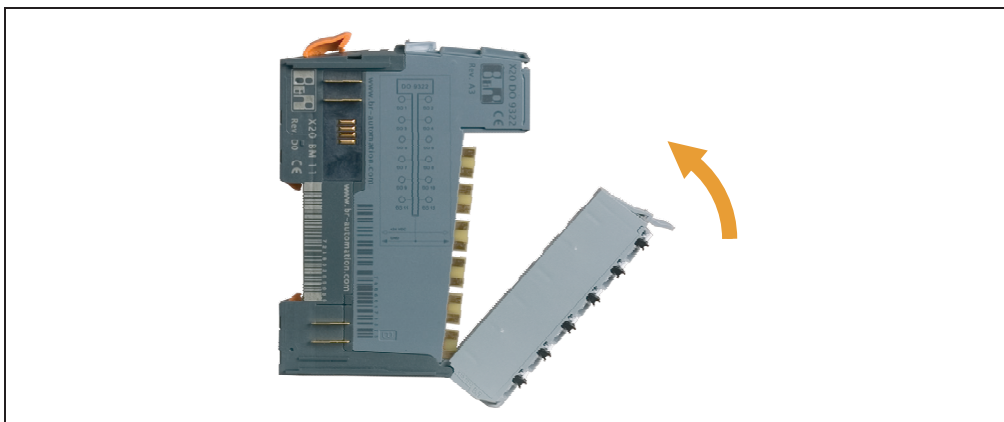


Figure 220: Rotate the terminal block up into place

- 5) Secure the terminal block in the electronic module by pushing in the terminal locking clip.

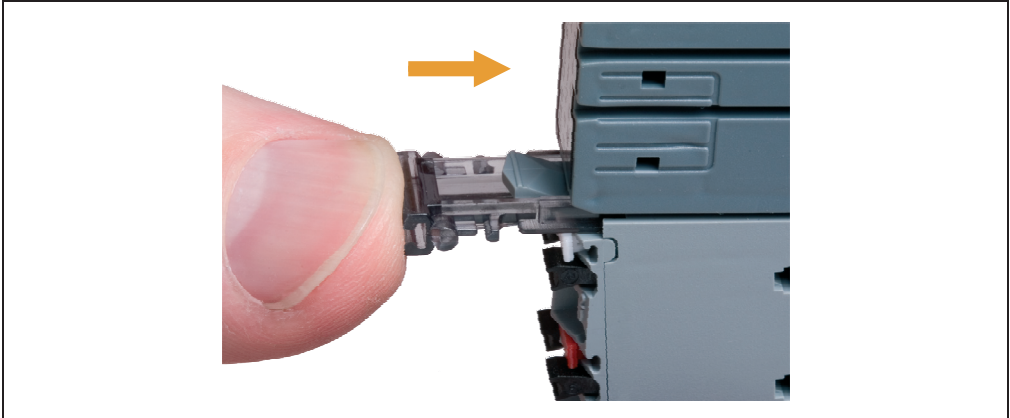


Figure 221: Secure the terminal block in the electronic module

- 6) Installed terminal locking clip.

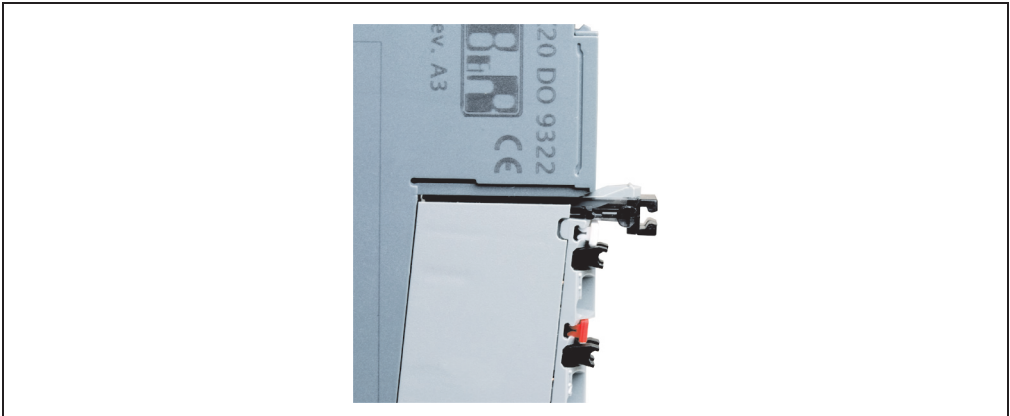


Figure 222: Installed terminal locking clip

- 7) To remove the terminal block, pull the terminal locking clip out again.

6.2 Plain text cover

Covers are available for X20 modules into which plain text legend strips can be inserted. The covers are attached to the terminal locking clips.

- 1) Hold the plain text cover at a 90° angle to the terminal locking clip.
- 2) Push the plain text cover into the terminal locking clip until it clicks into the slot on the clip.

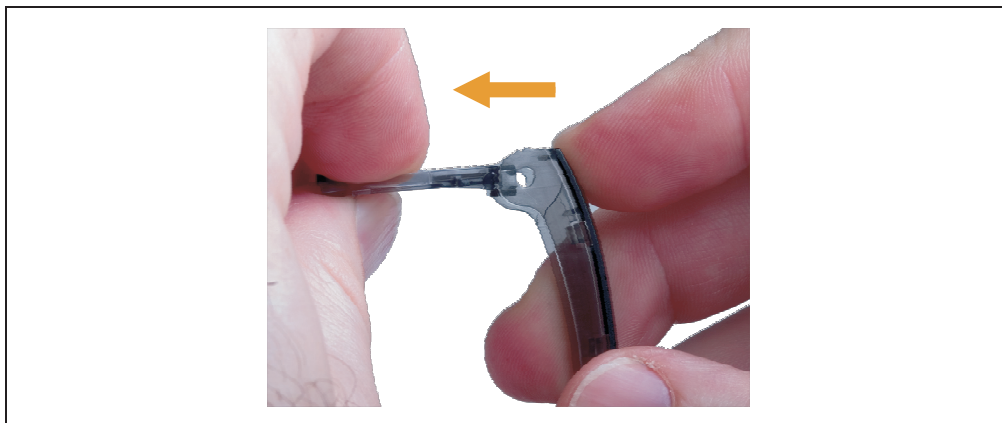
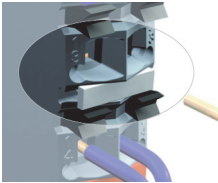
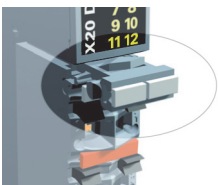
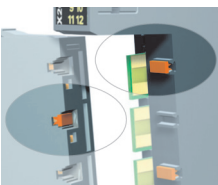


Figure 223: Attach plain text cover to terminal locking clip

7. Label tabs

The label tabs can serve the following purposes:

	Labeling the terminals		Labeling the module
	Labeling the terminal blocks		Coding the terminals

The labeling tool is needed to attach the label tabs.



Figure 224: Labeling tool

7.1 Labeling the terminals

This section explains how to label the terminals. The terminal blocks and the modules are labeled in a similar manner.

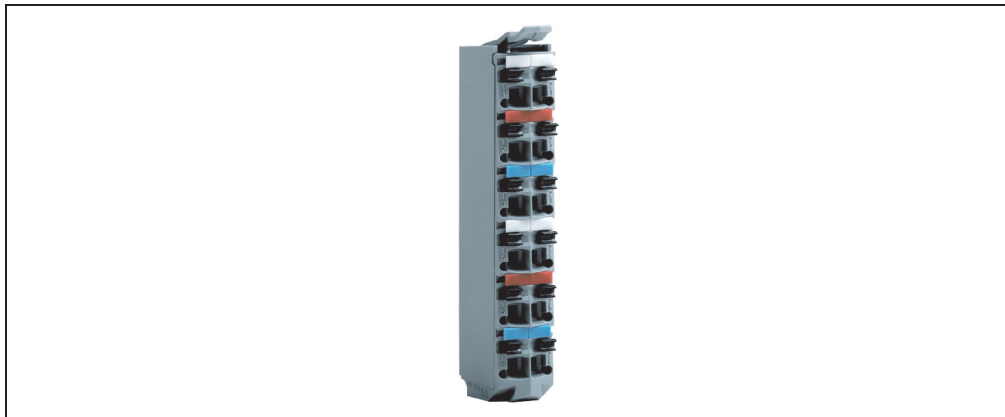


Figure 225: Terminal block with label tabs

- 1) Grip the desired label tabs with the double-width cutters of the labeling tool.

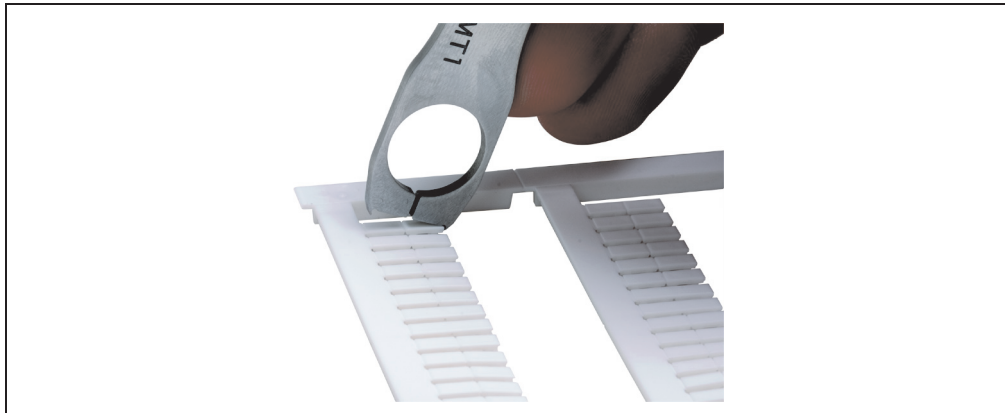


Figure 226: Grip desired label tabs with labeling tool

- 2) Press with the labeling tool to separate the labels.

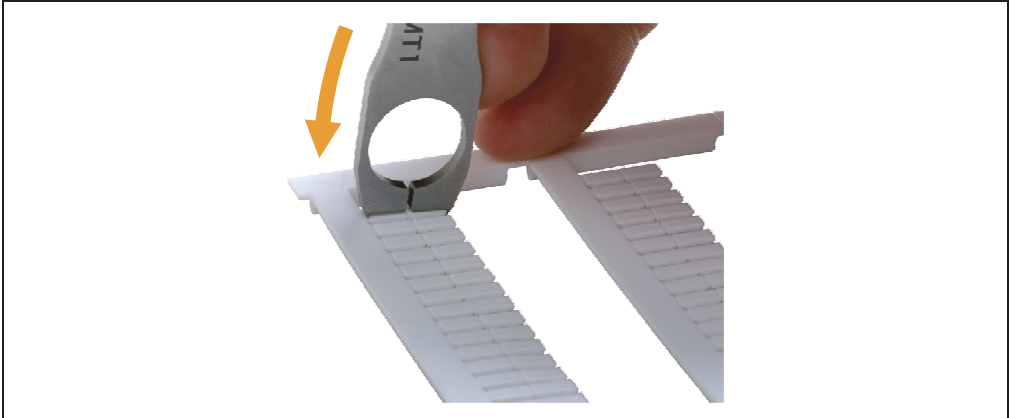


Figure 227: Separate labels with the labeling tool

- 3) Center the label tabs over the slot on the terminal block.

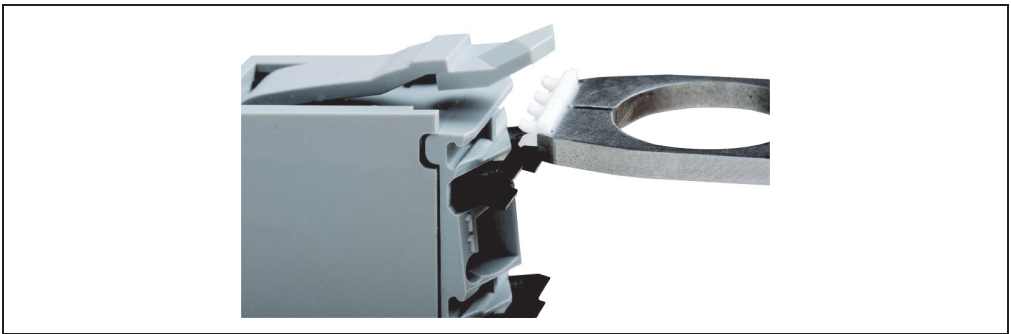


Figure 228: Center label tabs over slot

- 4) Hold the labeling tool at approximately an 80° angle to the terminal block.

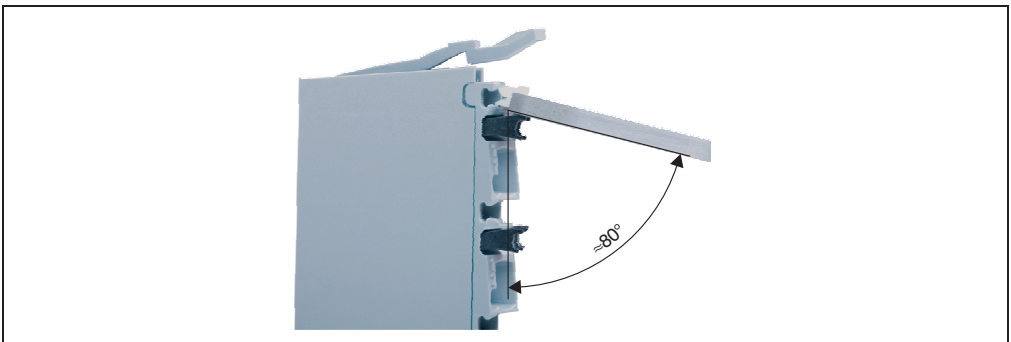


Figure 229: Hold the labeling tool at approximately 80° angle to the terminal block

- 5) Press with the labeling tool to insert the feet of the label tabs into the slot.
- 6) Inserted label.

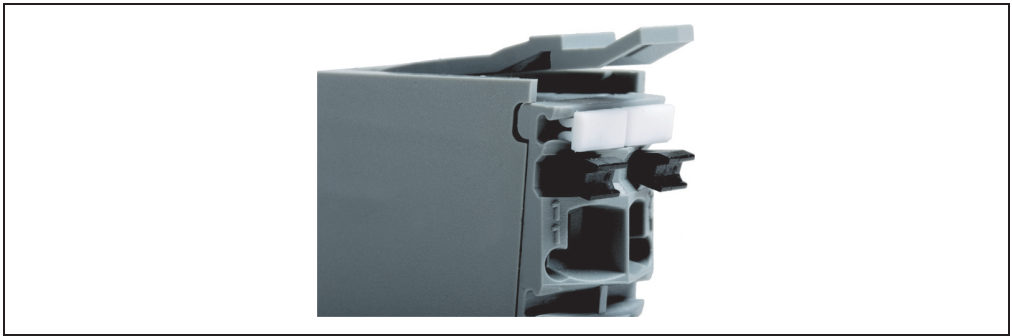


Figure 230: Inserted label

7.2 Coding the terminals

To prevent errors, the X20 terminal blocks can be coded. This helps prevent terminal blocks from being inserted in the wrong electronic module.

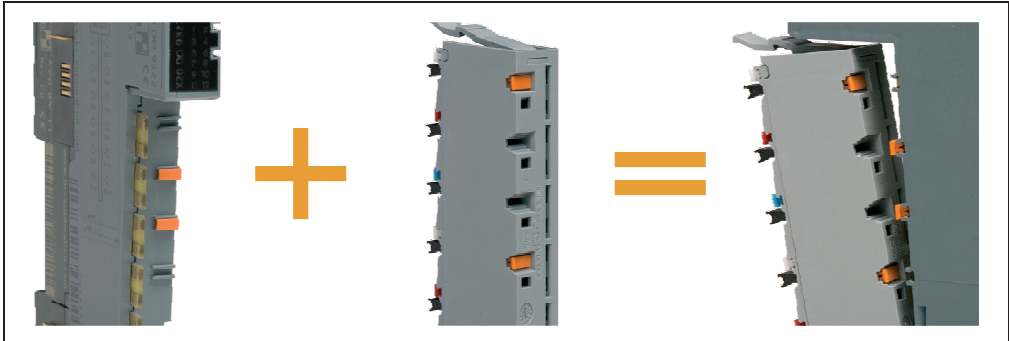


Figure 231: Terminal coding helps prevent errors from the start

- 1) Grip the desired label tab with the single-width cutters of the labeling tool (compare with section 7.1 "Labeling the terminals" on page 448).
- 2) Center the label tab over the slot on the electronic module.

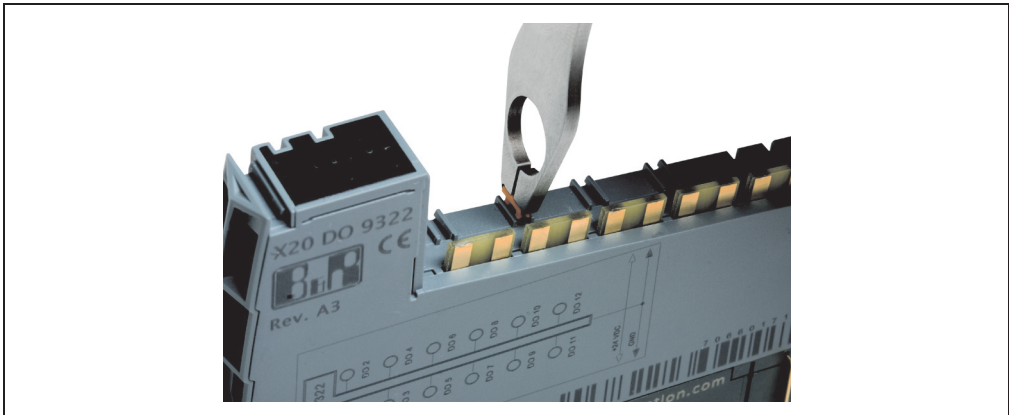


Figure 232: Center the label tab over the slot on the electronic module

- 3) Hold the labeling tool at a 90° angle to the electronic module and press to insert the label's feet into the slot.
- 4) Remove a label tab with the single-width cutter of the labeling tool.

- 5) Set the label tab in the slot on the back of the terminal block as shown.

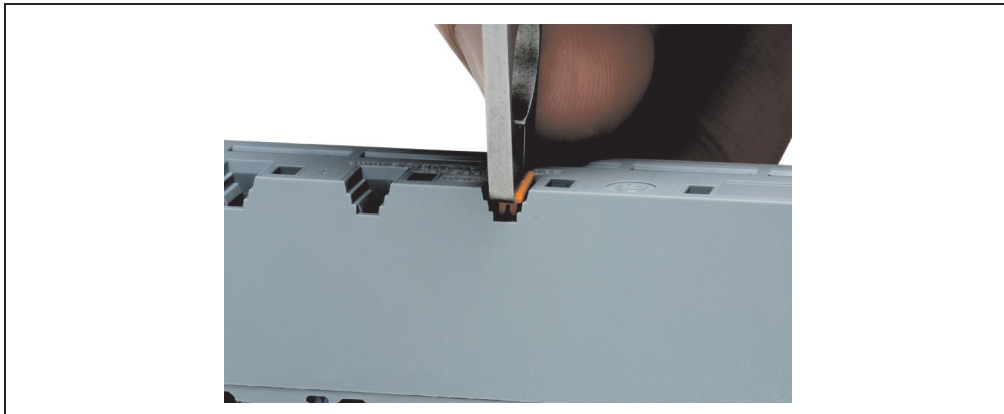


Figure 233: Set the label tab in the slot on the back of the terminal block

- 6) Use the labeling tool to push the left feet of the label into the slot.

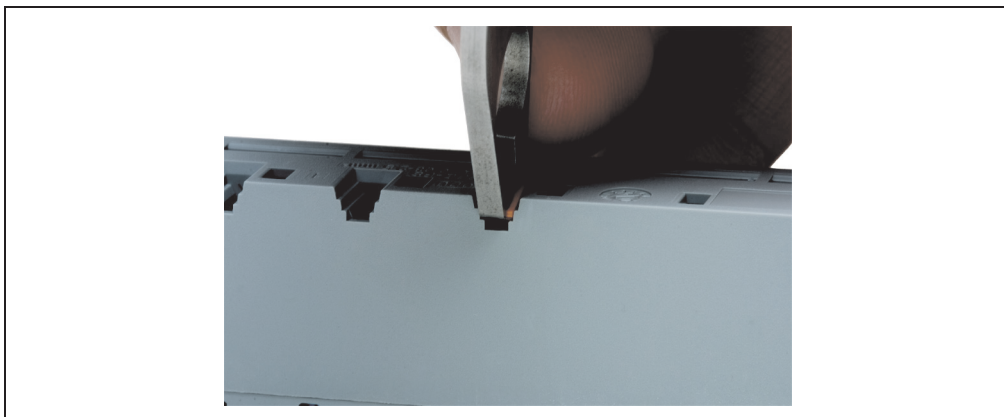


Figure 234: Press left feet of the label into the holes

7) With the labeling tool, press the right feet of the label into the slot.

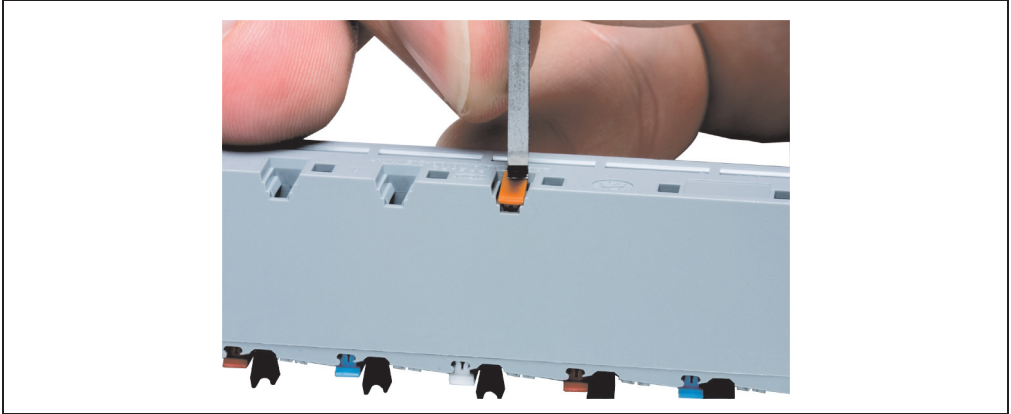


Figure 235: Press right feet of the label into the slot

8) Inserted label for terminal coding.

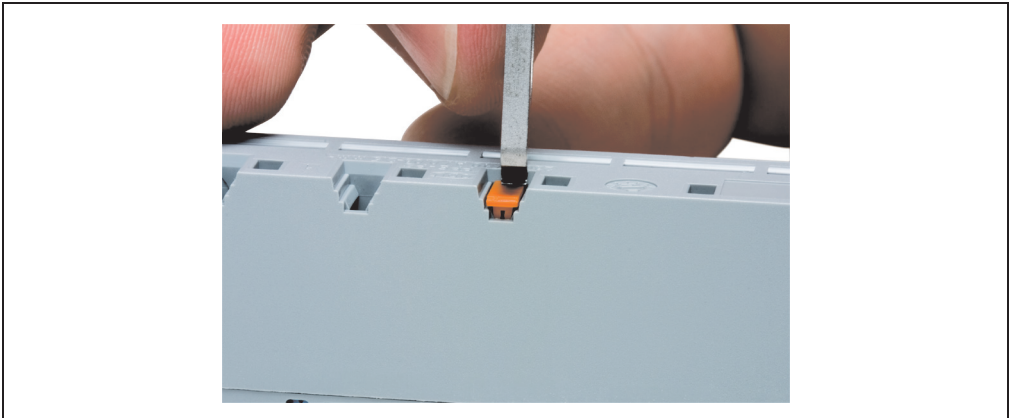


Figure 236: Inserted label for terminal coding

Chapter 7 • Standards and certifications

1. Standards and limits used

The product standard EN 61131-2 is generally valid for B&R industrial products (identical with IEC 61131-2). The following standards provide a detailed definition of proper functionality in a typical industrial environment (charged with electromagnetic energy):

Standard	Description
EN 61000-6-4	Electromagnetic compatibility (EMC); Generic standard - emission standard Part 2: Industrial environment (EN 50081-2 will be replaced by IEC 61000-6-4)
IEC/CISPR 11	Industrial, scientific and medical high-frequency device radio disturbances limits and measuring procedures
EN 61000-6-2	Electromagnetic compatibility (EMC) - Generic standard, emission standard Part 2: Industrial environment (EN 50082-2 will be replaced by IEC 61000-6-2)
EN 61131-2 Edition 2	Programmable logic controllers Part 2: Equipment requirements and tests
UL 508	Industrial control equipment, (UL = Underwriters Laboratories)
EN 60204-1	Safety of machinery - electrical equipment on machines Part 1: General requirements
EN 60529	IP20 protection

Table 230: Overview of standards

1.1 Tests, limits, and test subject behavior

Criteria	Description
A	Uninterrupted operation during the test.
B	Brief interruption during the test allowed.

Table 231: The criteria A and B describe the test subject behavior

EN 61000-4-2, electrostatic discharge	
	IEC 61131-2 Ed. 2
Contact discharge to powder-coated and bare metal parts	±4 kV, criteria B
Air discharge to plastic parts	±8 kV, criteria B

Table 232: Limits for electrostatic discharge

EN 61000-4-3, electromagnetic fields	
Housing, completely wired	80 MHz - 2 GHz, 10 V/m, criteria A 80% amplitude modulation at 1 kHz

Table 233: Limits for electromagnetic fields

EN 61000-4-4, burst, fast transients	
	IEC 61131-2 Edition 2
Power supplies and AC inputs/outputs	2 kV, 1 min, criteria B
Data cable	1 kV, 1 min, criteria B
Digital inputs/outputs	1 kV, 1 min, criteria B
Analog inputs/outputs	1 kV, 1 min, criteria B

Table 234: Limits for fast transients

EN 61000-4-5, surge		
	Common mode, unsymmetrical	Differential mode, symmetrical
AC power supplies	2 kV (12 Ω), criteria B	1 kV (2 Ω), criteria B
DC power supplies	1 kV (12 Ω), criteria B	0.5 kV (2 Ω), criteria B
Digital and analog I/O, AC, unshielded AC auxiliary voltage outputs for sensors, etc.	2 kV (42 Ω), criteria B	1 kV (42 Ω), criteria B
Digital and analog I/O, DC, unshielded Data lines, unshielded DC auxiliary voltage outputs for sensors, etc.	0.5 kV (42 Ω), criteria B	0.5 kV (42 Ω), criteria B
All shielded lines	1 kV (2 Ω), criteria B	---

Table 235: Limits for surge

EN 61000-4-6, conducted disturbances (radio frequency)	
Network I/O	150 kHz - 80 MHz, 10 V, criteria A
Signal connections >10 m	80% amplitude modulation at 1 kHz
Functional ground connection	

Table 236: Limits for conducted disturbances (radio frequency)

EN 55011 (IEC/CISPR 11, modified), emission	
	IEC 61131-2 Edition 2
30 MHz - 230 MHz	40 dB (μV/m), quasi-peak value, measured at distance of 10 m
230 MHz - 1 GHz	47 dB (μV/m), quasi-peak value, measured at distance of 10 m

Table 237: Emission

EN 60664-1, pollution degree
Pollution degree 2: non-conductive material

Table 238: Degree of pollution

EN 60068-2-6, vibration		
Frequency range [Hz]	Continuous	Periodic
$5 \leq f < 9$	1.75 mm amplitude	3.5 mm amplitude
$9 \leq f \leq 150$	0.5 g constant acceleration	1 g constant acceleration
$f > 150$	Not defined	Not defined

Table 239: Limit values for vibration

EN 60068-2-27, shock
15 g over 11 ms, half sinus wave, 3 shocks per axis and direction (total 18 shocks)

Table 240: Limit values for shock

2. International standards

B&R products and services comply with all applicable standards. They are international standards from organizations such as ISO, IEC and CENELEC, as well as national standards from organizations such as UL, CSA, FCC, VDE, ÖVE, etc. We give special consideration to the reliability of our products in an industrial environment.

Certifications	
USA and Canada 	All important B&R products are tested and listed by Underwriters Laboratories and are checked quarterly by a UL inspector. This certification is valid in the USA and Canada and makes it considerably easier to license your machines and systems in these markets.
Europe 	All harmonized EN standards for the valid guidelines are met.
Russian Federation 	GOST-R certification is available for the export of all B&R products in the Russian Federation. The X20 System is in preparation to be sent in for certification.

Appendix A • Abbreviations

1. General information

Abbreviations appear throughout the User's Manual, for example in data tables or descriptions of pin assignments.

2. Overview

Abbreviation	Stands for	Description
NC	Normally closed	A relay contact. Also known as an "opener".
	Not connected	Used in the description of pin assignments if a terminal or pin is not connected to a module.
ND	Not defined	In data tables, this stands for a value that has not been defined. Because a cable manufacturer does not provide certain technical data, for example.
NO	Normally open	A relay contact. Also known as an "closer".
TBD	To be defined	Used in the technical data tables when certain pieces of information are not yet available. The value will be provided later.

Table 241: Abbreviations used in the User's Manual

Figure 1:	Three basic elements result in one module: Terminal block – Electronic module – Bus module	31
Figure 2:	X20 modules are divided into three parts to guarantee the simplest usability	33
Figure 3:	Features of the X20 System.....	34
Figure 4:	X2X Link - universal backplane based on twisted copper cables.....	35
Figure 5:	Expansion of existing control systems using standard fieldbuses and the X20 System	36
Figure 6:	X20, X67, XV - variations of a single system	37
Figure 7:	Easy mounting on and removal from the mounting rail	39
Figure 8:	Visual diagnostics directly on the module using LED displays.....	40
Figure 9:	X20 System configurator for fieldbus connection	45
Figure 10:	X20 System configurator for connection to X2X Link backplane.....	46
Figure 11:	X20 System - I/O module dimensions	47
Figure 12:	X20 System - bus controller dimensions	48
Figure 13:	X20 System - installation guidelines.....	50
Figure 14:	Stress relief using cable ties.....	51
Figure 15:	Slots through which the cable ties are fed.....	51
Figure 16:	The bus module replaces the rack in the X20 System	52
Figure 17:	Simple implementation of various potential groups.....	53
Figure 18:	Protection of various potential groups	55
Figure 19:	Protection for supply feed via bus transmitter	56
Figure 20:	Example of expanded X2X Link supply	58
Figure 21:	Example of redundant X2X Link supply	59
Figure 22:	BM01 - potential control	72
Figure 23:	BM11 - potential control	74
Figure 24:	The four parts of a bus controller - fieldbus interface, base module, supply module, terminal block	77
Figure 25:	BC0043 - operational and connection elements	82
Figure 26:	BC0043 - terminating resistance	83
Figure 27:	BC0043 - number switches for node numbers and transfer rates.....	84
Figure 28:	BC0053 - operational and connection elements	93
Figure 29:	BC0053 - terminating resistance	94
Figure 30:	BC0053 - address switches.....	95
Figure 31:	BC0063 - operational and connection elements	101
Figure 32:	BC0063 - number switch	102
Figure 33:	BC0073 - operational and connection elements	108
Figure 34:	BC0073 - terminating resistance	109
Figure 35:	BC0073 - number switches for node numbers and transfer rates.....	110
Figure 36:	BC0083 - operational and connection elements	116
Figure 37:	BC0083 - station number switches.....	116
Figure 38:	BC0083 - RJ45 ports.....	117
Figure 39:	BB80 - potential control	120
Figure 40:	PS9400 - pin assignments	124
Figure 41:	PS9400 - connection example with two separate supplies	125
Figure 42:	PS9400 - connection example with a supply and jumper.....	125
Figure 43:	BR9300 - pin assignments	132
Figure 44:	BR9300 - connection example with two separate supplies	133

Figure index

Figure 45:	BR9300 - connection example with a supply and jumper	133
Figure 46:	BT9100 - pin assignments.....	139
Figure 47:	BT9100 - connection example - no feed for internal I/O supply	140
Figure 48:	BT9100 - connection example - with feed for internal I/O supply	140
Figure 49:	BT9100 - X2X Link supply according to system.....	141
Figure 50:	PS2100 - pin assignments	147
Figure 51:	PS2100 - connection example	147
Figure 52:	PS3300 - pin assignments	152
Figure 53:	PS3300 - connection example with two separate supplies	153
Figure 54:	PS3300 - connection example with a supply and jumper.....	153
Figure 55:	Input filter.....	156
Figure 56:	DI2371 - pin assignments.....	161
Figure 57:	DI2371 - connection example.....	161
Figure 58:	DI2371 - input circuit diagram	162
Figure 59:	DI2377 - pin assignments.....	168
Figure 60:	DI2377 - connection example.....	168
Figure 61:	DI2377 - input circuit diagram	169
Figure 62:	DI4371 - pin assignments.....	175
Figure 63:	DI4371 - connection example.....	175
Figure 64:	DI4371 - input circuit diagram	176
Figure 65:	DI6371 - pin assignments.....	181
Figure 66:	DI6371 - connection example.....	181
Figure 67:	DI6371 - input circuit diagram	182
Figure 68:	DI6372 - pin assignments.....	187
Figure 69:	DI6372 - connection example.....	187
Figure 70:	DI6372 - input circuit diagram	188
Figure 71:	DI9371 - pin assignments.....	193
Figure 72:	DI9371 - connection example.....	193
Figure 73:	DI9371 - input circuit diagram	194
Figure 74:	DI9372 - pin assignments.....	199
Figure 75:	DI9372 - connection example.....	199
Figure 76:	DI9372 - input circuit diagram	200
Figure 77:	DO2322 - pin assignments.....	207
Figure 78:	DO2322 - connection example.....	207
Figure 79:	DO2322 - output circuit diagram	208
Figure 80:	DO2322 - switching inductive loads	208
Figure 81:	DO4322 - pin assignments.....	214
Figure 82:	DO4322 - connection example.....	214
Figure 83:	DO4322 - output circuit diagram	215
Figure 84:	DO4322 - switching inductive loads	215
Figure 85:	DO4332 - pin assignments.....	221
Figure 86:	DO4332 - connection example.....	221
Figure 87:	DO4332 - output circuit diagram	222
Figure 88:	DO4332 - switching inductive loads	222
Figure 89:	DO6321 - pin assignments.....	228
Figure 90:	DO6321 - connection example.....	228
Figure 91:	DO6321 - output circuit diagram	229

Figure 92:	DO6321 - switching inductive loads	229
Figure 93:	DO6322 - pin assignments	234
Figure 94:	DO6322 - connection example	234
Figure 95:	DO6322 - output circuit diagram	235
Figure 96:	DO6322 - switching inductive loads	235
Figure 97:	DO8332 - pin assignments	240
Figure 98:	DO8332 - connection example	240
Figure 99:	DO8332 - output circuit diagram	241
Figure 100:	DO8332 - switching inductive loads	241
Figure 101:	DO8332 - derating when 3 or 4 channels are operated with 2 A	242
Figure 102:	DO9321 - pin assignment	249
Figure 103:	DO9321 - connection example	249
Figure 104:	DO9321 - output circuit diagram	250
Figure 105:	DO9321 - switching inductive loads	250
Figure 106:	DO9322 - pin assignments	255
Figure 107:	DO9322 - connection example	255
Figure 108:	DO9322 - output circuit diagram	256
Figure 109:	DO9322 - switching inductive loads	256
Figure 110:	AI2622 - pin assignments	269
Figure 111:	AI2622 - connection example	269
Figure 112:	AI2622 - input circuit diagram	270
Figure 113:	AI2632 - pin assignments	276
Figure 114:	AI2632 - connection example	277
Figure 115:	AI2632 - input circuit diagram	278
Figure 116:	AI4622 - pin assignments	283
Figure 117:	AI4622 - connection example	283
Figure 118:	AI4622 - input circuit diagram	284
Figure 119:	AI4632 - pin assignments	290
Figure 120:	AI4632 - connection example	291
Figure 121:	AI4632 - input circuit diagram	292
Figure 122:	AO2622 - pin assignments	297
Figure 123:	AO2622 - connection example	297
Figure 124:	AO2622 - output circuit diagram	298
Figure 125:	AO2632 - pin assignments	303
Figure 126:	AO2632 - connection example	303
Figure 127:	AO2632 - output circuit diagram	304
Figure 128:	AO4622 - pin assignments	309
Figure 129:	AO4622 - connection example	309
Figure 130:	AO4622 - output circuit diagram	310
Figure 131:	AO4632 - pin assignments	315
Figure 132:	AO4632 - connection example	315
Figure 133:	AO4632 - output circuit diagram	316
Figure 134:	AT2222 - pin assignments	323
Figure 135:	AT2222 - connection example	323
Figure 136:	AT2222 - input circuit diagram - 2-wire connection	324
Figure 137:	AT2222 - input circuit diagram - 3-wire connection	324
Figure 138:	AT4222 - pin assignments	333

Figure index

Figure 139: AT4222 - connection example.....	333
Figure 140: AT4222 - input circuit diagram - 2-wire connection	334
Figure 141: AT4222 - input circuit diagram - 3-wire connection	334
Figure 142: AT2402 - pin assignments.....	343
Figure 143: AT2402 - connection example.....	343
Figure 144: AT2402 - input circuit diagram	344
Figure 145: AT6402 - pin assignments.....	353
Figure 146: AT6402 - connection example.....	353
Figure 147: AT6402 - input circuit diagram	354
Figure 148: DC1196 - pin assignments	363
Figure 149: DC1196 - connection example	363
Figure 150: DC1196 - input circuit diagram - counter inputs	364
Figure 151: DC1196 - input circuit diagram - standard inputs	364
Figure 152: DC1198 - pin assignments	369
Figure 153: DC1198 - connection example	369
Figure 154: DC1198 - input circuit diagram - counter inputs	370
Figure 155: DC1198 - input circuit diagram - standard inputs	370
Figure 156: DC1198 - output circuit diagram.....	371
Figure 157: DC1396 - pin assignments	375
Figure 158: DC1396 - connection example	375
Figure 159: DC1396 - input circuit diagram - counter inputs	376
Figure 160: DC1396 - input circuit diagram - standard input.....	376
Figure 161: DC1398 - pin assignments	381
Figure 162: DC1398 - connection example	381
Figure 163: DC1398 - input circuit diagram - counter inputs	382
Figure 164: DC1398 - input circuit diagram - standard input.....	382
Figure 165: DC1398 - output circuit diagram.....	382
Figure 166: DC2396 - pin assignments	386
Figure 167: DC2396 - connection example	386
Figure 168: DC2396 - input circuit diagram - counter inputs	387
Figure 169: DC2396 - input circuit diagram - standard input.....	387
Figure 170: DC2398 - pin assignments	393
Figure 171: DC2398 - connection example	393
Figure 172: DC2398 - input circuit diagram - counter inputs	394
Figure 173: DC2398 - input circuit diagram - standard inputs	394
Figure 174: DC2398 - output circuit diagram.....	394
Figure 175: DC2395 - pin assignments	400
Figure 176: DC2395 - connection example	400
Figure 177: DC2395 - input circuit diagram.....	402
Figure 178: DC2395 - output circuit diagram.....	402
Figure 179: DC2395 - switching inductive loads	403
Figure 180: DC4395 - pin assignments	410
Figure 181: DC4395 - connection example	410
Figure 182: DC4395 - input circuit diagram.....	412
Figure 183: DC4395 - output circuit diagram.....	412
Figure 184: DC4395 - switching inductive loads	413
Figure 185: ZF0000 - pin assignments.....	418

Figure 186: ZF0000 - connection example.....	418
Figure 187: Additional equipment for X20 modules.....	419
Figure 188: Insert electronic module in the guides on the bus module	426
Figure 189: Push the electronic module and the bus module flush together.....	427
Figure 190: Hang the bottom edge of the terminal block in its place on the bus module	427
Figure 191: Rotate the terminal block up into place	427
Figure 192: If the latch does not catch, the lever must be pushed up	428
Figure 193: Slide the right module forward into the guides on the left bus module.....	428
Figure 194: Insert the right locking plate into guides on the bus module from the front	429
Figure 195: Lay the left locking plate on the left module and insert it in the guides	429
Figure 196: Push the locking lever all the way up to open the locking mechanism.....	430
Figure 197: Insert the next bus module in the guides of the previously mounted bus module	430
Figure 198: Insert the electronic module in the guides on the leftmost bus module.....	431
Figure 199: Push the electronic module and the bus module flush together.....	431
Figure 200: Insert right locking plate into the guides on the bus module from the front	432
Figure 201: Hang the terminal block in its place on the leftmost bus module	432
Figure 202: Rotate the terminal block up into place	433
Figure 203: If the latch does not catch, the lever must be pushed up	433
Figure 204: Lay the left locking plate on the left module and insert it in the guides	434
Figure 205: Push the locking lever all the way up to open the locking mechanism.....	435
Figure 206: Push the locking lever all the way up to open the locking mechanism.....	436
Figure 207: Push the locking lever all the way up to open the locking mechanism.....	436
Figure 208: Remove the terminal block from the module to the left	437
Figure 209: Module block removed from the mounting rail	438
Figure 210: Hang the bottom edge of the terminal block in its place on the bus module	438
Figure 211: Rotate the terminal block up into place	439
Figure 212: If the latch does not catch, the lever must be pushed up	439
Figure 213: Unplug the terminal block on the rightmost module	440
Figure 214: Remove the electronic module.....	440
Figure 215: Unlock the right locking plate with a screwdriver.....	441
Figure 216: Installing the additional locking clip	442
Figure 217: Set the terminal locking clip on the terminal block locking lever	443
Figure 218: Install terminal locking clip on terminal block	443
Figure 219: Hang the bottom edge of the terminal block in its place on the bus module	444
Figure 220: Rotate the terminal block up into place	444
Figure 221: Secure the terminal block in the electronic module.....	445
Figure 222: Installed terminal locking clip.....	445
Figure 223: Attach plain text cover to terminal locking clip.....	446
Figure 224: Labeling tool.....	447
Figure 225: Terminal block with label tabs	448
Figure 226: Grip desired label tabs with labeling tool.....	448
Figure 227: Separate labels with the labeling tool.....	449
Figure 228: Center label tabs over slot.....	449
Figure 229: Hold the labeling tool at approximately 80° angle to the terminal block	449
Figure 230: Inserted label.....	450
Figure 231: Terminal coding helps prevent errors from the start.....	451

Figure index

Figure 232: Center the label tab over the slot on the electronic module	451
Figure 233: Set the label tab in the slot on the back of the terminal block	452
Figure 234: Press left feet of the label into the holes	452
Figure 235: Press right feet of the label into the slot	453
Figure 236: Inserted label for terminal coding	453

Table 1:	Manual history	25
Table 2:	Organization of safety notices	29
Table 3:	Definition of terms.....	29
Table 4:	X20 modules - overview for calculating the power output table	60
Table 5:	Overview 1 - The modules are arranged in ascending alphabetical order	65
Table 6:	Overview 2 - The modules are arranged according to their module group	67
Table 7:	Overview of bus modules	70
Table 8:	BM01 - order data.....	71
Table 9:	BM01 - technical data.....	71
Table 10:	BM11 - order data.....	73
Table 11:	BM11 - technical data.....	73
Table 12:	TB06/TB12 - order data.....	75
Table 13:	TB06/TB12 - technical data	76
Table 14:	Bus controller overview	78
Table 15:	BC0043 - order data	79
Table 16:	BC0043 - technical data	80
Table 17:	BC0043 - additional technical data.....	81
Table 18:	BC0043 - status indicators	81
Table 19:	BC0043 - pin assignments - CAN interface	82
Table 20:	BC0043 - node numbers and transfer rates	84
Table 21:	BC0043 - transfer rate search table	85
Table 22:	BC0043 - possible fixed transfer rates	86
Table 23:	BC0053 - order data.....	89
Table 24:	BC0053 - technical data	89
Table 25:	BC0053 - additional technical data.....	90
Table 26:	BC0053 - status indicators	91
Table 27:	BC0053 - pin assignments - DeviceNet interface.....	93
Table 28:	BC0053 - MAC ID.....	95
Table 29:	BC0053 - special functions.....	95
Table 30:	BC0053 - transfer rate search table	96
Table 31:	BC0063 - order data	99
Table 32:	BC0063 - technical data	99
Table 33:	BC0063 - additional technical data.....	100
Table 34:	BC0063 - status indicators	101
Table 35:	BC0063 - pin assignments - Profibus DP interface	102
Table 36:	BC0063 - node numbers	102
Table 37:	BC0063 - transfer rates supported by the bus controller	103
Table 38:	BC0063 - status LED description	104
Table 39:	BC0073 - order data.....	105
Table 40:	BC0073 - technical data	106
Table 41:	BC0073 - additional technical data.....	107
Table 42:	BC0073 - status indicators	107
Table 43:	BC0073 - pin assignments - CAN interface.....	108
Table 44:	BC0073 - node numbers and transfer rates	110
Table 45:	BC0073 - transfer rate search table	111
Table 46:	BC0083 - order data.....	112
Table 47:	BC0083 - technical data	113

Table index

Table 48:	BC0083 - additional technical data.....	114
Table 49:	BC0083 - status indicators	114
Table 50:	BC0083 - Status/Error LED lit red: LED indicates error.....	115
Table 51:	BC0083 - Status/Error LED lit green: LED indicates operation	115
Table 52:	BC0083 - station numbers	116
Table 53:	BC0083 - pin assignments - RJ45 port.....	117
Table 54:	Overview of bus controller system modules	118
Table 55:	BB80 - order data	119
Table 56:	BB80 - technical data	119
Table 57:	PS9400 - order data	121
Table 58:	PS9400 - technical data	122
Table 59:	PS9400 - additional technical data	123
Table 60:	PS9400 - status indicators.....	124
Table 61:	Overview of bus transmitters and bus receivers.....	128
Table 62:	BR9300 - order data	129
Table 63:	BR9300 - technical data	129
Table 64:	BR9300 - additional technical data.....	131
Table 65:	BR9300 - status indicators	131
Table 66:	BT9100 - order data	136
Table 67:	BT9100 - technical data.....	137
Table 68:	BT9100 - additional technical data	138
Table 69:	BT9100 - status indicators.....	138
Table 70:	BT9100 - system dependant X2X Link supply.....	141
Table 71:	Overview of supply modules.....	143
Table 72:	PS2100 - order data	144
Table 73:	PS2100 - technical data	145
Table 74:	PS2100 - additional technical data	146
Table 75:	PS2100 - status indicators.....	146
Table 76:	PS3300 - order data	149
Table 77:	PS3300 - technical data	150
Table 78:	PS3300 - additional technical data	151
Table 79:	PS3300 - status indicators.....	151
Table 80:	Digital input module overview.....	157
Table 81:	Digital input module with additional functions.....	157
Table 82:	DI2371 - order data	158
Table 83:	DI2371 - technical data.....	159
Table 84:	DI2371 - additional technical data	160
Table 85:	DI2371 - status indicators.....	160
Table 86:	DI2377 - order data	164
Table 87:	DI2377 - technical data.....	165
Table 88:	DI2377 - additional technical data	166
Table 89:	DI2377 - status indicators.....	167
Table 90:	DI4371 - order data	172
Table 91:	DI4371 - technical data.....	173
Table 92:	DI4371 - additional technical data	174
Table 93:	DI4371 - status indicators.....	174
Table 94:	DI6371 - order data	178

Table 95:	DI6371 - technical data.....	179
Table 96:	DI6371 - additional technical data	180
Table 97:	DI6371 - status indicators.....	180
Table 98:	DI6372 - order data	184
Table 99:	DI6372 - technical data.....	185
Table 100:	DI6372 - additional technical data	186
Table 101:	DI6372 - status indicators.....	186
Table 102:	DI9371 - order data	190
Table 103:	DI9371 - technical data.....	191
Table 104:	DI9371 - additional technical data	192
Table 105:	DI9371 - status indicators.....	192
Table 106:	DI9372 - order data	196
Table 107:	DI9372 - technical data.....	197
Table 108:	DI9372 - additional technical data	198
Table 109:	DI9372 - status indicators.....	198
Table 110:	Overview 1 of digital output modules.....	202
Table 111:	Overview 2 of digital output modules.....	202
Table 112:	DO2322 - order data.....	203
Table 113:	DO2322 - technical data.....	204
Table 114:	DO2322 - additional technical data	205
Table 115:	DO2322 - status indicators.....	206
Table 116:	DO4322 - order data.....	210
Table 117:	DO4322 - technical data.....	211
Table 118:	DO4322 - additional technical data	212
Table 119:	DO4322 - status indicators.....	213
Table 120:	DO4332 - order data.....	217
Table 121:	DO4332 - technical data.....	218
Table 122:	DO4332 - additional technical data	219
Table 123:	DO4332 - status indicators.....	220
Table 124:	DO4332 - operation with 2 A	223
Table 125:	DO6321 - order data.....	225
Table 126:	DO6321 - technical data.....	226
Table 127:	DO6321 - additional technical data	227
Table 128:	DO6321 - status indicators.....	227
Table 129:	DO6322 - order data.....	231
Table 130:	DO6322 - technical data.....	232
Table 131:	DO6322 - additional technical data	233
Table 132:	DO6322 - status indicators.....	233
Table 133:	DO8332 - order data.....	237
Table 134:	DO8332 - technical data.....	238
Table 135:	DO8332 - additional technical data	239
Table 136:	DO8332 - status indicators.....	239
Table 137:	DO8332 - operation with 2 A	242
Table 138:	DO9321 - order data.....	246
Table 139:	DO9321 - technical data.....	247
Table 140:	DO9321 - additional technical data	248
Table 141:	DO9321 - status indicators.....	248

Table index

Table 142: DO9322 - order data.....	252
Table 143: DO9322 - technical data.....	253
Table 144: DO9322 - additional technical data	254
Table 145: DO9322 - status indicators.....	254
Table 146: Overview of analog input modules	258
Table 147: AI2622 - order data.....	265
Table 148: AI2622 - technical data.....	266
Table 149: AI2622 - additional technical data	267
Table 150: AI2622 - status indicators.....	268
Table 151: AI2632 - order data.....	273
Table 152: AI2632 - technical data.....	274
Table 153: AI2632 - additional technical data	275
Table 154: AI2632 - status indicators.....	276
Table 155: AI4622 - order data.....	279
Table 156: AI4622 - technical data.....	280
Table 157: AI4622 - additional technical data	281
Table 158: AI4622 - status indicators.....	282
Table 159: AI4632 - order data.....	287
Table 160: AI4632 - technical data.....	288
Table 161: AI4632 - additional technical data	289
Table 162: AI4632 - status indicators.....	290
Table 163: Overview of analog output modules	293
Table 164: AO2622 - order data.....	294
Table 165: AO2622 - technical data.....	295
Table 166: AO2622 - additional technical data.....	296
Table 167: AO2622 - status indicators	296
Table 168: AO2632 - order data.....	300
Table 169: AO2632 - technical data.....	300
Table 170: AO2632 - additional technical data.....	302
Table 171: AO2632 - status indicators	302
Table 172: AO4622 - order data.....	306
Table 173: AO4622 - technical data	307
Table 174: AO4622 - additional technical data.....	308
Table 175: AO4622 - status indicators	308
Table 176: AO4632 - order data.....	312
Table 177: AO4632 - technical data	312
Table 178: AO4632 - additional technical data.....	314
Table 179: AO4632 - status indicators	314
Table 180: Overview of temperature modules.....	318
Table 181: AT2222 - order data	319
Table 182: AT2222 - technical data.....	320
Table 183: AT2222 - additional technical data	321
Table 184: AT2222 - status indicators.....	322
Table 185: AT4222 - order data	329
Table 186: AT4222 - technical data.....	330
Table 187: AT4322 - additional technical data	331
Table 188: AT4222 - status indicators.....	332

Table 189: AT2402 - order data	339
Table 190: AT2402 - technical data.....	340
Table 191: AT2402 - additional technical data	341
Table 192: AT2402 - status indicators.....	342
Table 193: AT6402 - order data	349
Table 194: AT6402 - technical data.....	350
Table 195: AT6402 - additional technical data	351
Table 196: AT6402 - status indicators.....	352
Table 197: Overview of counter modules	359
Table 198: DC1196 - order data.....	360
Table 199: DC1196 - technical data	360
Table 200: DC1196 - additional technical data.....	362
Table 201: DC1196 - status indicators	362
Table 202: DC1198 - order data.....	366
Table 203: DC1198 - technical data	366
Table 204: DC1198 - additional technical data.....	368
Table 205: DC1198 - status indicator.....	368
Table 206: DC1396 - order data.....	372
Table 207: DC1396 - technical data	372
Table 208: DC1396 - additional technical data.....	374
Table 209: DC1396 - status indicators	374
Table 210: DC1398 - order data.....	378
Table 211: DC1398 - technical data	378
Table 212: DC1398 - additional technical data.....	380
Table 213: DC1398 - status indicators	380
Table 214: DC2396 - order data.....	383
Table 215: DC2396 - technical data	383
Table 216: DC2396 - additional technical data.....	385
Table 217: DC2396 - status indicators	385
Table 218: DC2398 - order data.....	390
Table 219: DC2398 - technical data	390
Table 220: DC2398 - additional technical data.....	392
Table 221: DC2398 - status indicators	392
Table 222: DC2395 - order data.....	395
Table 223: DC2395 - technical data	396
Table 224: DC2395 - additional technical data.....	397
Table 225: DC2395 - status indicators	399
Table 226: DC4395 - order data.....	405
Table 227: DC4395 - technical data	406
Table 228: DC4395 - additional technical data.....	407
Table 229: DC4395 - status indicators	409
Table 230: ZF0000 - order data.....	416
Table 231: ZF0000 - technical data.....	417
Table 232: Two of the several methods for assembling an X20 System.....	426
Table 233: Overview of standards.....	455
Table 234: The criteria A and B describe the test subject behavior	456
Table 235: Limits for electrostatic discharge	456

Table index

Table 236: Limits for electromagnetic fields	456
Table 237: Limits for fast transients.....	456
Table 238: Limits for surge	456
Table 239: Limits for conducted disturbances (radio frequency).....	457
Table 240: Emission	457
Table 241: Degree of pollution	457
Table 242: Limit values for vibration.....	457
Table 243: Limit values for shock	457
Table 244: Abbreviations used in the User's Manual	459

Numbers

1, 2, or 3-wire connections	43
2D representation	48
3D representation	48

A

Abbreviations	459
Accessories	419
Additional locking clip	420
AI2622	265
AI2632	273
AI4622	279
AI4632	287
Analog input modules	258
AI2622	265
AI2632	273
AI4622	279
AI4632	287
Analog output modules	293
AO2622	294
AO2632	300
AO4622	306
AO4632	312
AO2622	294
AO2632	300
AO4622	306
AO4632	312
AT2222	319
AT2402	339
AT4222	329
AT6402	349

B

B&R industrial products	
Limits used	455
Safety notices	26
Backplane	35
BB80	119
BC0043	79
BC0053	88
BC0063	98
BC0073	105
BC0083	112
BM01	71

BM11	73
BR9300	129
BT9100	136
Bus controller	36, 44, 77
BC0043	79
BC0053	88
BC0063	98
BC0073	105
BC0083	112
System modules	118
Bus modules	52, 70
BB80	119
BM01	71
BM11	73
Bus receiver	54
BR9300	129
Bus receivers	128
Bus supply	53
Bus transmitter	54, 56
Bus transmitters	128
BT9100	136

C

Cable ties	51
CAD support	48, 49
Certifications	458
Connection type	43
Construction support	48, 49
Counter modules	359
DC1196	360
DC1198	366
DC1396	372
DC1398	378
DC2395	395
DC2396	383
DC2398	390
DC4395	405
Cover holder	420

D

DC1196	360
DC1198	366
DC1396	372
DC1398	378

DC2395	395
DC2396	383
DC2398	390
DC4395	405
Decentralized backplane	35
Definition of terms	29
DI2371	158
DI2377	164
DI4371	172
DI6371	178
DI6372	184
DI9371	190
DI9372	196
Diagnostics	40
Digital input modules	156
DI2371	158
DI2377	164
DI4371	172
DI6371	178
DI6372	184
DI9371	190
DI9372	196
Digital output modules	202
DO2322	203
DO4322	210
DO4332	217
DO6321	225
DO6322	231
DO8332	237
DO9321	246
DO9322	252
Dimensions	47
DO2322	203
DO4322	210
DO4332	217
DO6321	225
DO6322	231
DO8332	237
DO9321	246
DO9322	252
Dummy Module ZF0000	416

E

ECAD macros	48, 49
Electrical configuration	47
Electrostatic discharge, protection	27
Embedded parameter chip	40
ESD note	28
Expanded X2X Link supply	57

F

Fieldbus technology	36, 44
Fuse protection	55

I

Ink cartridge	421
Installation	47, 50
Mechanical design	425

L

Label tabs	422
Labeling tool	423
Legend	459
Definition of terms	29
Safety notices	29
Limit values	455
Limits used	455
Locking plate	421

M

Manual history	25
Mechanical configuration	47
Mechanical design	425
Module overview	65
Mounting rail	50

O

Operated by PLC	28
Organization of safety notices	29

P

Plain text cover	420
Potential groups	53, 55
Power output table	60
Power supply	
Expanded X2X Link supply	57
Redundant X2X Link supply	57
Power supply modules	54, 143
PS2100	144
PS3300	149
PS9400	121
Printer	49, 421
Printer cartridge	421
Protection against electrostatic discharges	27
PS2100	144
PS3300	149
PS9400	121

R

Redundant X2X Link supply	57
---------------------------------	----

S

Safety notices	26
Intended use	26
Operation	28
Organization	29
Transport and storage	28
SG3 and SG4	29
Standards	455, 458
STEP data	48
Storage	28
Stress relief	51
Supply modules	54, 143
PS2100	144
PS3300	149
PS9400	121
System characteristics	31
System configurator	44
System Generation 3 and 4	29

T

TB06	75
TB12	75
Temperature modules	318
AT2222	319
AT2402	339
AT4222	329
AT6402	349
Terminal blocks	75
TB06	75
TB12	75
Terminal labeling	49, 422
Terminal locking clip	420
The power supply design	52
Transport	28

V

Valve manifold control	37
------------------------------	----

X

X20 System	31
X20 System configuration	44
X2X Link	35, 57
X67 System	37

Z

ZF0000	416
--------------	-----

X

X20AC0AX1	420	X20AT4222	329
X20AC0AX1.0100	420	X20AT6402	349
X20AC0CB1	421	X20BB80	119
X20AC0LB1.0100	420	X20BC0043	79
X20AC0M01	422	X20BC0053	89
X20AC0M01.0010	422	X20BC0063	99
X20AC0M02	422	X20BC0073	105
X20AC0M02.0010	422	X20BC0083	112
X20AC0M03	422	X20BM01	71
X20AC0M03.0010	422	X20BM11	73
X20AC0M04	422	X20BR9300	129
X20AC0M04.0010	422	X20BT9100	136
X20AC0M11	422	X20DC1196	360
X20AC0M11.0010	422	X20DC1198	366
X20AC0M12	422	X20DC1396	372
X20AC0M12.0010	422	X20DC1398	378
X20AC0M13	422	X20DC2395	395
X20AC0M13.0010	422	X20DC2396	383
X20AC0M14	422	X20DC2398	390
X20AC0M14.0010	422	X20DC4395	405
X20AC0MT1	423	X20DI2371	158
X20AC0PR1	421	X20DI2377	164
X20AC0SC1	420	X20DI4371	172
X20AC0SC1.0100	420	X20DI6371	178
X20AC0SH1	420	X20DI6372	184
X20AC0SH1.0100	420	X20DI9371	190
X20AC0SL1	421	X20DI9372	196
X20AC0SL1.0010	421	X20DO2322	203
X20AC0SR1	421	X20DO4322	210
X20AC0SR1.0010	421	X20DO4332	217
X20AI2622	265	X20DO6321	225
X20AI2632	273	X20DO6322	231
X20AI4622	279	X20DO8332	237
X20AI4632	287	X20DO9321	246
X20AO2622	294	X20DO9322	252
X20AO2632	300	X20PS2100	144
X20AO4622	306	X20PS3300	149
X20AO4632	312	X20PS9400	121
X20AT2222	319	X20TB06	75
X20AT2402	339	X20TB12	75
		X20ZF0000	416

